

L13 Price of Anarchy

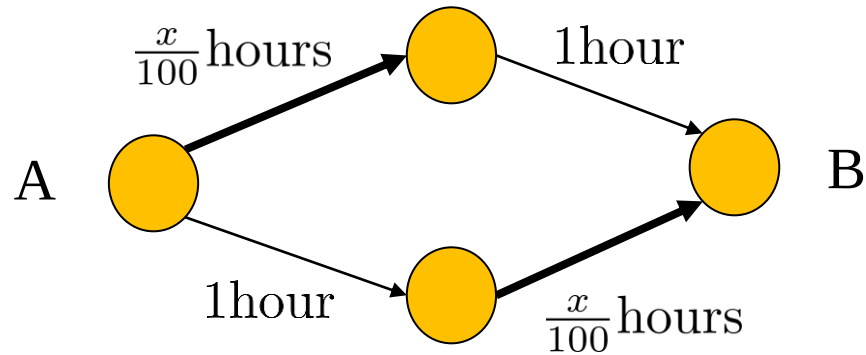
CS 280 Algorithmic Game Theory

Ioannis Panageas

Price of Anarchy

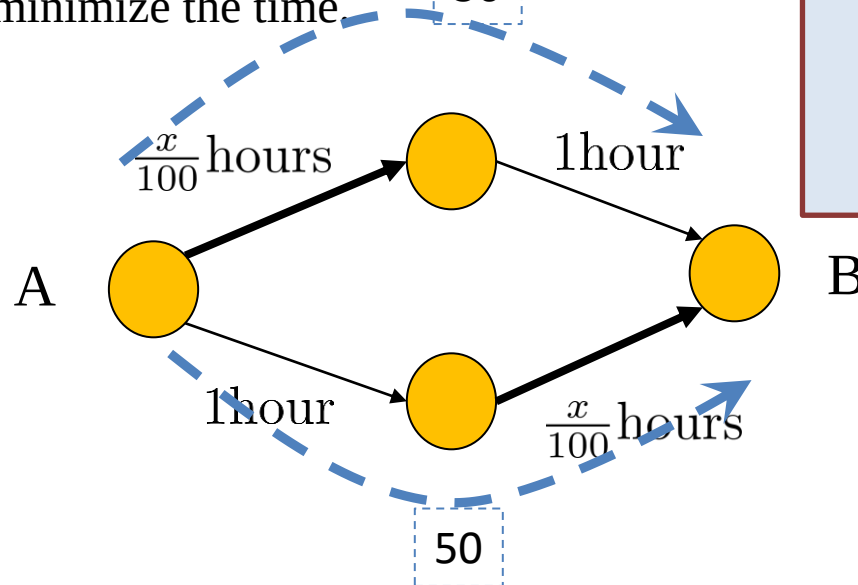
Suppose 100 drivers commute from A to B.

Drivers want to minimize the time.



Price of Anarchy

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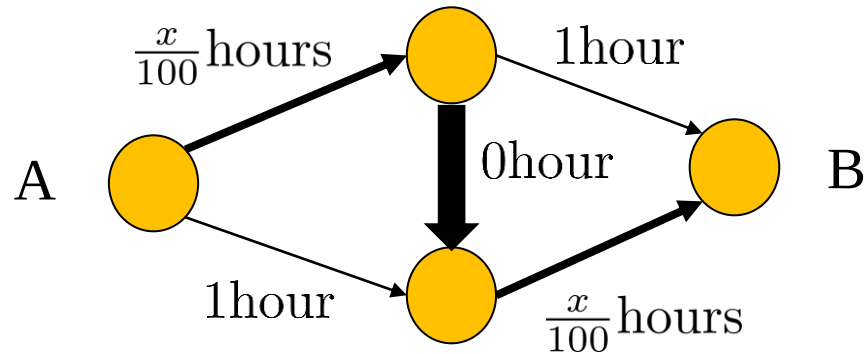


Delay is 1.5 hours for everybody at the unique Nash equilibrium.

Price of Anarchy

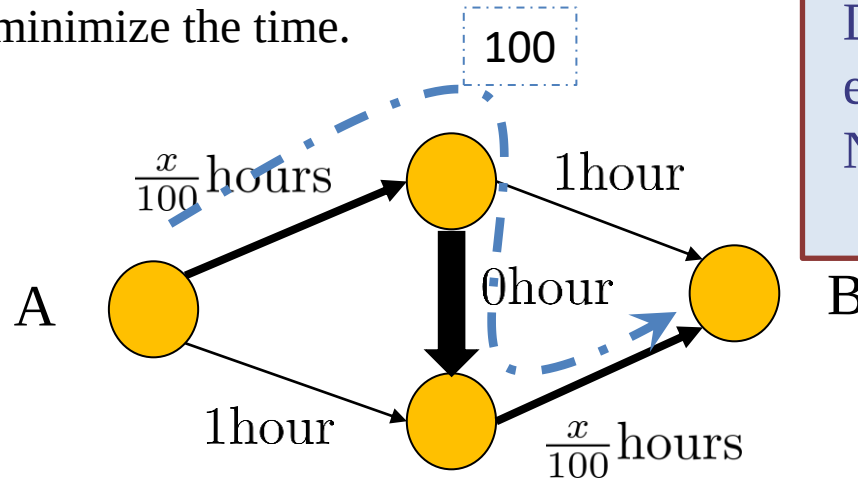
Suppose 100 drivers commute from A to B.
Drivers want to minimize the time.

Question: What if we **add** a new link?



Price of Anarchy

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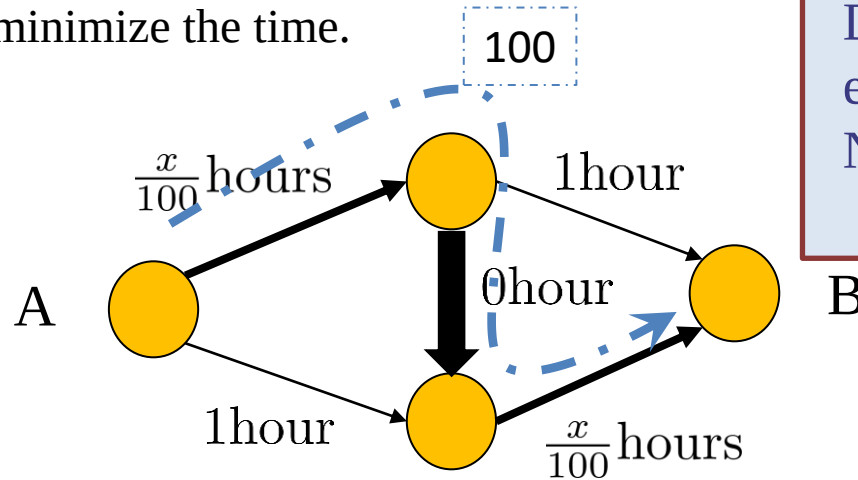


Delay is now 2 hours for everybody at the unique Nash equilibrium.
Braess's paradox

Adding a fast link is not always a good idea!

Price of Anarchy

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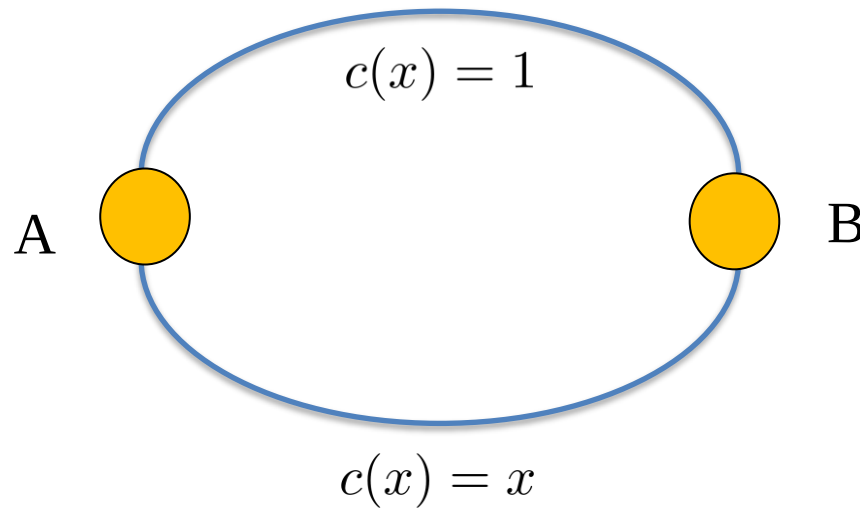
Adding a fast link is not always a good idea!

PoA = $\frac{\text{performance of worst case NE}}{\text{optimal performance if agents do not decide on their own}}$
Price of Anarchy (Koutsoupias, Papadimitriou 99').

4/3!!

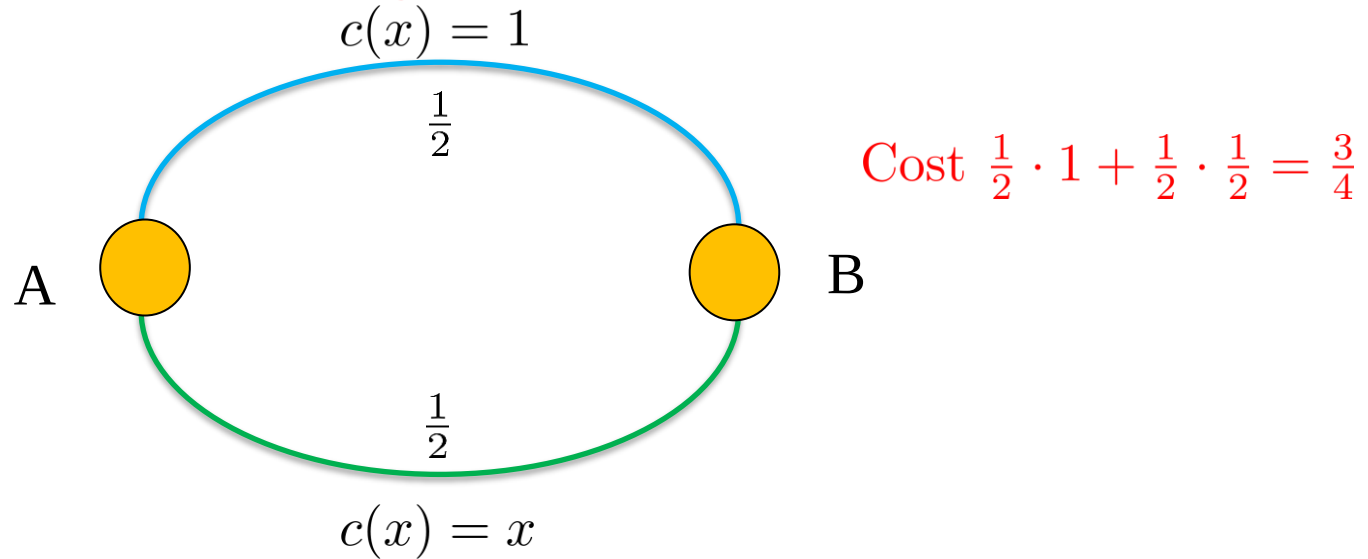
Non-atomic selfish routing

Example: Simpler example. **Pigou network.**



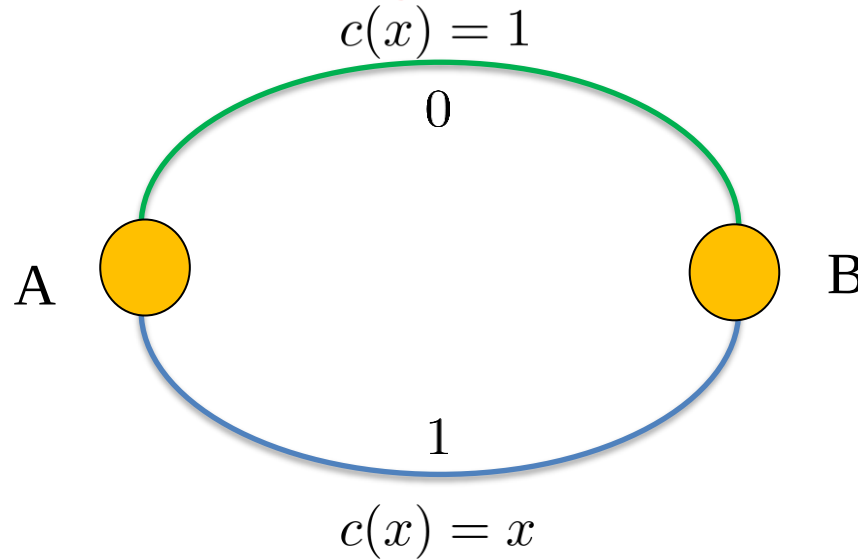
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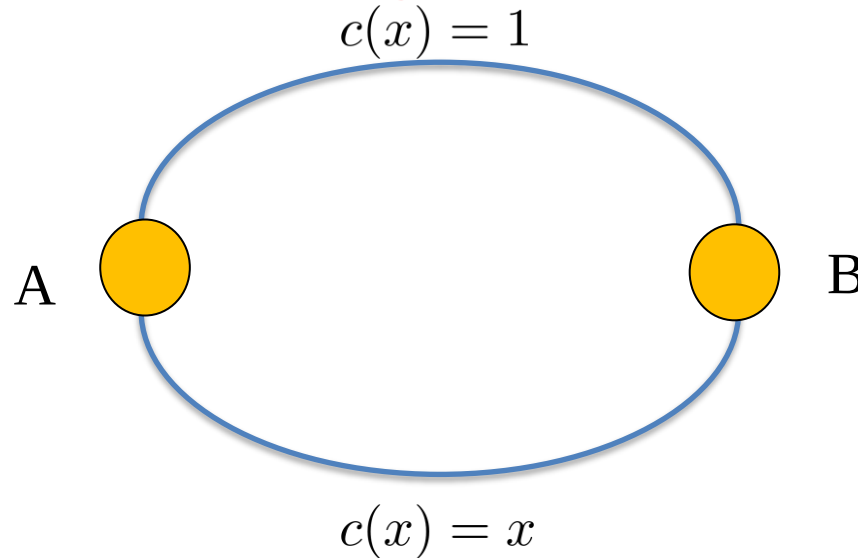
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Cost 1 and **PoA** = $\frac{4}{3}$

Non-atomic selfish routing

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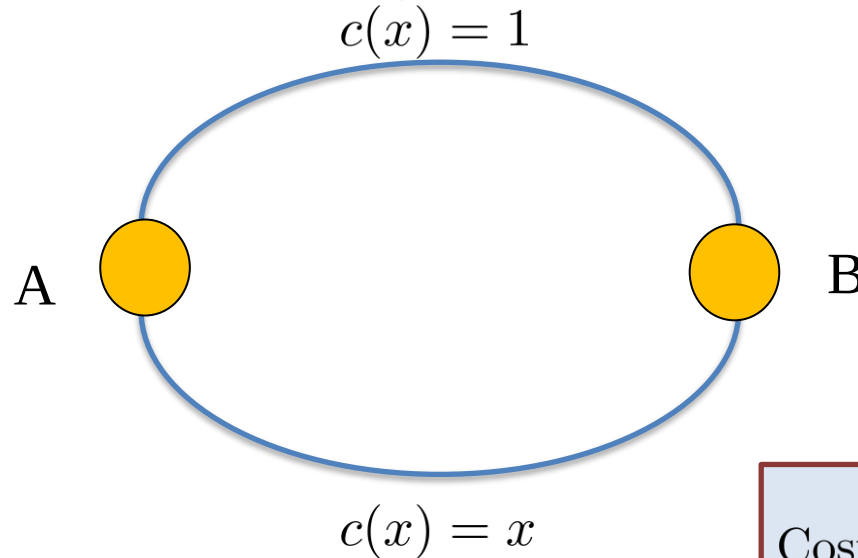


A **non-atomic selfish routing** game is defined by:

- Graph $G(V, E)$.
- Source destination pairs $(s_1, t_1), \dots, (s_k, t_k)$.
- r_i traffic from $s_i \rightarrow t_i$.
- $c_e(\cdot) \geq 0$ cost function of edge e , continuous and non-decreasing.
- Flow is an equilibrium if all traffic is routed on **cheapest paths**.

Non-atomic selfish routing

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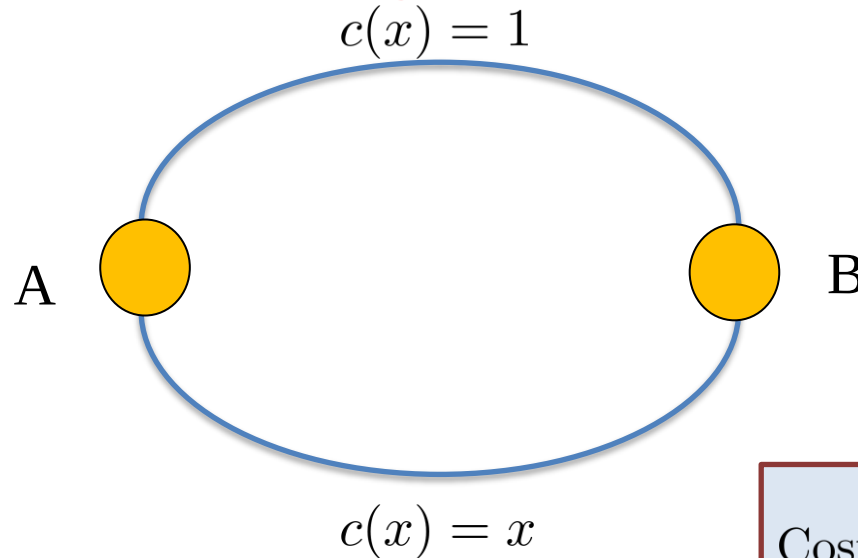
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$$\begin{aligned}\text{Cost of path: } c_p(f) &= \sum_{e \in p} c_e(f) \\ \text{Social Cost} &:= \sum_p f_p c_p(f)\end{aligned}$$

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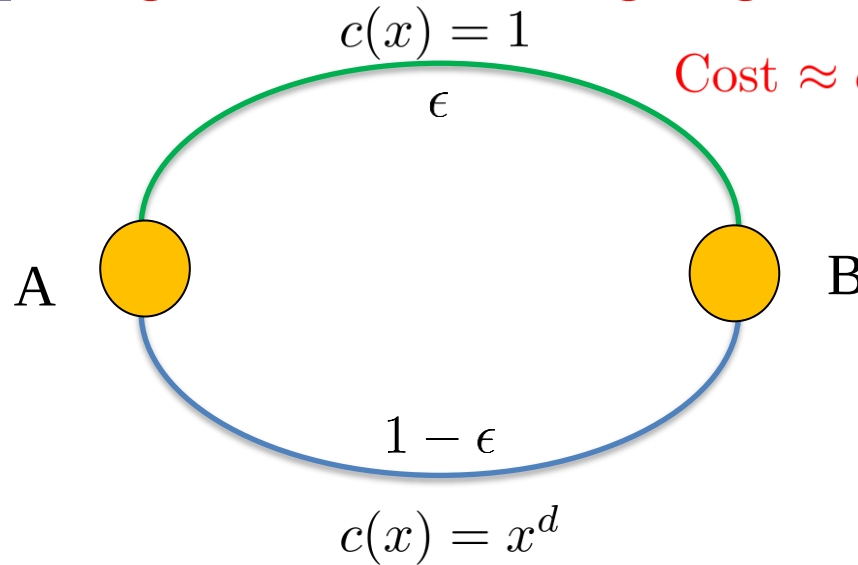
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Cost of path: $c_p(f) = \sum_{e \in p} c_e(f)$
Social Cost $:= \sum_p f_p c_p(f)$

Remark: Equilibrium flow exists and is unique!

Non-atomic selfish routing

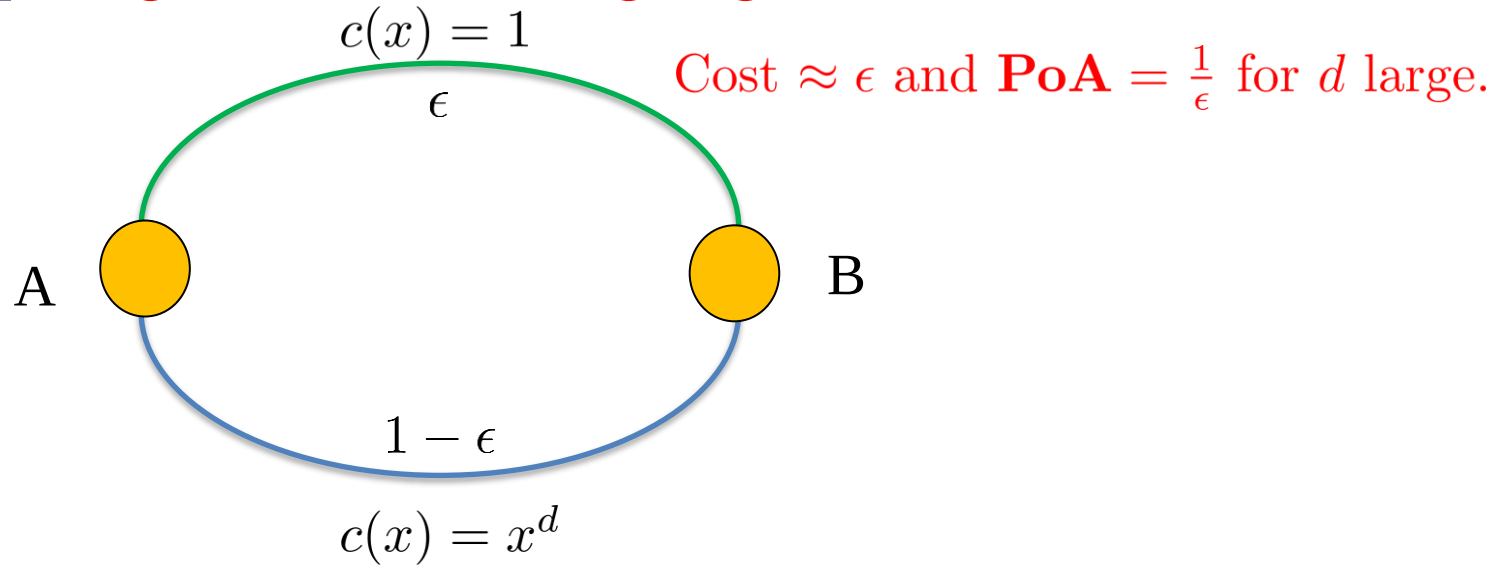
A bad Example. **Pigou network with large degree d .**



Cost $\approx \epsilon$ and **PoA** $= \frac{1}{\epsilon}$ for d large.

Non-atomic selfish routing

A bad Example. **Pigou network with large degree d .**



Questions:

1. When is PoA small (bounded)?
2. Can we find bounds on PoA for specific classes of cost functions?

Price of Anarchy in Non-atomic selfish routing with *Linear* costs

Definition (**Linear costs**). *Linear costs are of the form $c_e(x) = a_e \cdot x + b_e$.*

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$$\sum_e f_e^* c_e(f_e^*) \leq \sum_e f_e c_e(f_e^*).$$

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Observe that

f^* equilibrium flow $\Rightarrow$ if $f_p^* > 0$ then $c_p(f^*) \leq c_{p'}(f^*)$ for all paths p' .

Price of Anarchy in Non-atomic selfish routing with *Linear* costs

Proof cont. Therefore all paths p so that $f_p^* > 0$ have same cost say L .

Hence $\sum_p f_p^* c_p(f^*) = L \cdot F$ where $F = \sum_p f_p^*$ is the total flow.

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Hence $\sum_p f_p^* c_p(f^*) = L \cdot F$ where $F = \sum_p f_p^*$ is the total flow.

Since $c_p(f^*) \geq L$ we conclude

$$\sum_p f_p c_p(f^*) \geq L \sum_p f_p = L \cdot F$$

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Combining the above

$$\sum_e f_e c_e(f^*) = \sum_p f_p c_p(f^*) \geq L \cdot F = \sum_p f_p^* c_p(f^*) = \sum_e f_e^* c_e(f^*)$$

Price of Anarchy in Non-atomic selfish routing with *Linear* costs

$$\sum_e f_e c_e(f^*) \geq \sum_e f_e^* c_e(f^*).$$

Proof cont. We get that

$$\sum_e f_e^* c_e(f^*) \leq \sum_e f_e c_e(f) + \sum_e f_e (c_e(f^*) - c_e(f))$$

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We also have that

$$\sum_e f_e (c_e(f^*) - c_e(f)) \leq \frac{1}{4} \sum_e f_e^* c_e(f^*)$$

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- Case $c_e(f^*) \geq c_e(f) \Rightarrow f_e^* \geq f_e$. Linear costs $\Rightarrow$ LHS = $a_e f_e (f_e^* - f_e)$ and RHS $\geq \frac{1}{4} a_e f_e^*{}^2$.

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- Case $c_e(f^*) < c_e(f) \Rightarrow f_e \leq f_e^*$. Since $xy - y^2 \leq \frac{x^2}{4} \Rightarrow \text{LHS} \leq \text{RHS}.$
- Case $c_e(f^*) \geq c_e(f) \Rightarrow f_e \geq f_e^*$. Linear costs $\Rightarrow \text{LHS} = a_e f_e (f_e^* - f_e)$ and $\text{RHS} \geq \frac{1}{4} a_e f_e^{*2}.$

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Or equivalently

$$\sum_e f_e^* c_e(f^*) \leq \frac{4}{3} \sum_e f_e c_e(f).$$

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Proof cont. $\forall$

Pigou example is tight!

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Theorem (Roughgarden 02', **PoA for polynomial costs**). For every network with *polynomial costs* with degree d :

$$\text{cost of Nash flow} \leq \Theta\left(\frac{d}{\log d}\right) \cdot \text{cost of optimal flow}.$$

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HW2

Price of Anarchy in Congestion Games

Theorem (Christodoulou-Koutsoupias, **PoA for linear costs**). *For every congestion game with **linear costs**:*

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Proof. Let l^* be a Nash equilibrium in which i uses path P_i and assume i deviates to path $\tilde{P}_i$. It holds (Variational Inequality)

$$\sum_{e \in P_i} c_e(l_e^*) \leq \sum_{e \in P_i \cap \tilde{P}_i} c_e(l_e^*) + \sum_{e \in \tilde{P}_i \setminus P_i} c_e(l_e^* + 1)$$

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Price of Anarchy in Congestion Games

$$\sum_{e \in P_i} c_e(l_e^*) \leq \sum_{e \in \tilde{P}_i} c_e(l_e^* + 1).$$

Proof cont. Consider any configuration $\tilde{l}$, where each agent j uses path $\tilde{P}_j$.
Summing for all agents i

$$\sum_{i \in [n]} \sum_{e \in P_i} c_e(l_e^*) \leq \sum_{i \in [n]} \sum_{e \in \tilde{P}_i} c_e(l_e^* + 1).$$

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Since $y(z + 1) \leq \frac{5}{3}y^2 + \frac{1}{3}z^2$ for naturals y, z
HW2

$$= \sum_e a_e \tilde{l}_e (l_e^* + 1) + b_e \tilde{l}_e.$$

$$\leq \sum_e a_e \left(\frac{5}{3} \tilde{l}_e^2 + \frac{1}{3} l_e^{*2} \right) + b_e \tilde{l}_e.$$

Price of Anarchy in Congestion Games

$$\sum_{i \in [n]} \sum_{e \in P_i} c_e(l_e^*) \leq \sum_e a_e \left(\frac{5}{3} \tilde{l}_e^2 + \frac{1}{3} l_e^{*2} \right) + b_e \tilde{l}_e$$

Proof cont. Observe that

$$\frac{5}{3} C(\tilde{l}) = \frac{5}{3} \sum_{i \in [n]} \sum_{e \in \tilde{P}_i} c_e(\tilde{l}_e) = \sum_e \frac{5}{3} a_e \tilde{l}_e^2 + \frac{5}{3} b_e \tilde{l}_e \geq \sum_e \frac{5}{3} a_e \tilde{l}_e^2 + b_e \tilde{l}_e$$

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Therefore

$$C(l^*) \leq \frac{5}{3} C(\tilde{l}) + \frac{1}{3} \sum_e a_e l_e^{*2}$$

Price of Anarchy in Congestion Games

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$$\begin{aligned} C(l^*) &\leq \frac{5}{3} C(\tilde{l}) + \frac{1}{3} \sum_e a_e l_e^{*2} \\ &\leq \frac{5}{3} C(\tilde{l}) + \frac{1}{3} C(l^*) \end{aligned}$$

Price of Anarchy in Congestion Games

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Proof cont. Ob

$$\frac{5}{3} C(\tilde{l}) =$$

$$C(l^*) \leq \frac{5}{2} C(\tilde{l}).$$

$$a_e \tilde{l}_e^2 + b_e \tilde{l}_e$$

Therefore

$$\begin{aligned} C(l^*) &\leq \frac{5}{3} C(\tilde{l}) + \frac{1}{3} \sum_e a_e l_e^{*2} \\ &\leq \frac{5}{3} C(\tilde{l}) + \frac{1}{3} C(l^*) \end{aligned}$$

Remark:

1. The above bound is tight!
2. For polynomial cost functions the PoA is exponential in d .

Price of Anarchy and Balls & Bins

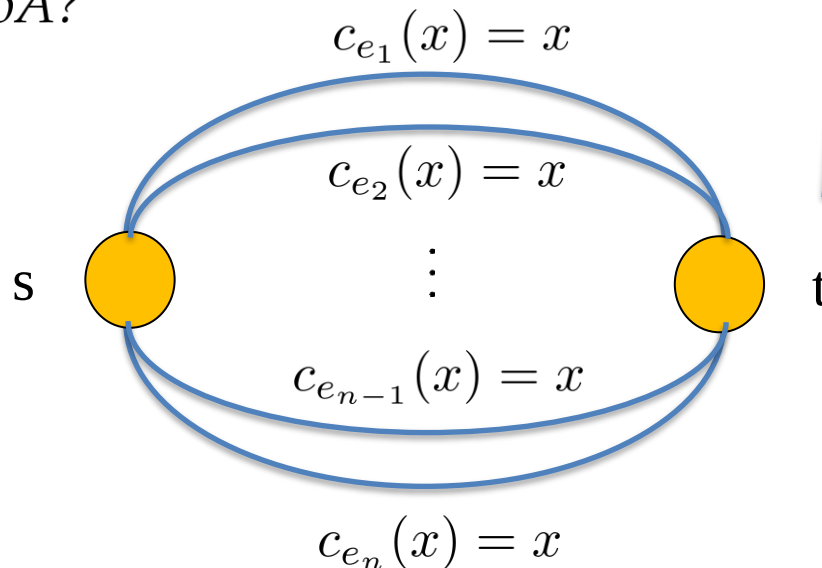
Definition (Balls and Bins). Consider

- set of n balls and n bins $\{e_1, \dots, e_n\}$.
- Each ball i chooses a bin j and pays the load of the bin j .
- Define social cost the *maximum load*.
- What is PoA? Is it $\frac{5}{2}$?

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Congestion game!

Price of Anarchy and Balls & Bins

Theorem (Koutsoupias-Papadimitriou, **PoA for balls & bins**). *The PoA is*

$$\Omega\left(\frac{\ln n}{\ln \ln n}\right).$$

Proof. We will use **second moment method**.

- Set every ball in a different bin. Hence optimal social cost is 1.
- Uniform $(\frac{1}{n}, \dots, \frac{1}{n})$ is a Nash Equilibrium (symmetry).

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- Set every ball in a different bin. Hence optimal social cost is 1.
- Uniform $(\frac{1}{n}, \dots, \frac{1}{n})$ is a Nash Equilibrium (symmetry).
- With high probability, we show that uniform gives max load $\Omega\left(\frac{\ln n}{\ln \ln n}\right)$, which implies the expected max load is $\Omega\left(\frac{\ln n}{\ln \ln n}\right)$.

Price of Anarchy and Balls & Bins

Theorem (Koutsoupias-Papadimitriou, **PoA for balls & bins**). *The PoA is*

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Proof. We will use **second moment method**.

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In general (HW2):

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Price of Anarchy and Balls & Bins

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thus $Pr[X = 0] \leq Pr[|X - E[X]| \geq E[X]] \leq \frac{Var[X]}{E^2[X]}.$

Price of Anarchy and Balls & Bins

Proof cont. $Pr[X = 0] \leq \frac{Var[X]}{E^2[X]}.$

From *negative correlation* we have that $Var[X] \leq \sum_i Var[X_i].$

Moreover $Var[X_i] = E[X_i^2] - E^2[X_i] \leq E[X_i^2] = E[X_i] \leq 1$

Price of Anarchy and Balls & Bins

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Therefore

$$Pr[X \geq 1] = 1 - Pr[X = 0] \geq 1 - \frac{n^{-1/3}}{e^2} \rightarrow 1.$$

Congestion Games

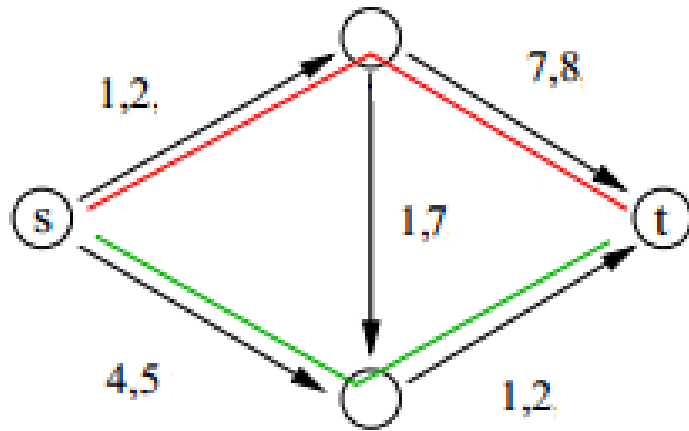
A **congestion game** is defined by:

- n set of players.
- E set of edges/facilities/ bins.
- $S_i \subset 2^E$ the set of strategies of player i .
- $c_e : \{1, \dots, n\} \rightarrow \mathbb{R}^+$ cost function of edge e .

For any $s = (s_1, \dots, s_n)$

- $l_e(s)$ number of players (load) that use edge e .
- $c_i(s) = \sum_{e \in s_i} c_e(l_e)$ the cost function of player i .

Congestion Games



For this game:

$n = \{1, 2\}$ (red, green)

E are the edges of the network.

S_i is all $s - t$ paths.

c_e on edges.

Remark: Defined by Rosenthal in 1973. Capture atomic routing **games**!